

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Pseudocode Writing Task

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO5 – Naturalization (independent application using OpenCV)P5	Assessment:	Assessment Tool 2 – Pseudocode Writing Task
Weightage:	25% of CIA	Total Marks:	100
CO4 Statement:	Develop computer vision applications using OpenCV for performing image and video processing tasks on images and videos		
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Section / Batch:	Section A	Date:	30 August 2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation	
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach	
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs	
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear	
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient	

Q15. OCR Preprocessing for Noisy Scanned Documents

Answer –

OCR (Optical Character Recognition) converts text present in scanned documents into machine-readable text. When scanned documents contain **noise, blur, uneven illumination, low contrast, skew, or broken characters**, directly applying OCR can produce incorrect results. Therefore, I would design a preprocessing pipeline to improve the quality of the document before recognition.

Proposed Pipeline

Input Image → Grayscale → Noise Removal → Contrast Enhancement → Thresholding → Deskewing → Morphological Processing → OCR

1. Grayscale Conversion

First, I convert the scanned document into grayscale.

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

This reduces the image from three color channels to a single intensity channel and simplifies further processing.

2. Noise Removal

I use **Median Filtering** to remove unwanted speckles and salt-and-pepper noise.

```
denoised = cv2.medianBlur(gray, 3)
```

This improves OCR accuracy by removing small unwanted pixels while preserving important character boundaries.

3. Contrast Enhancement

I apply histogram equalization to improve the visibility of faint text.

```
enhanced = cv2.equalizeHist(denoised)
```

This is useful when the scanned document has poor contrast between the text and background.

4. Binarization

Next, I convert the image into a binary image using adaptive thresholding.

```
binary = cv2.adaptiveThreshold(  
    enhanced, 255,  
    cv2.ADAPTIVE_THRESH_GAUSSIAN_C,  
    cv2.THRESH_BINARY,  
    11, 2  
)
```

Adaptive thresholding is suitable for scanned documents with **uneven lighting**, because the threshold is calculated locally.

5. Deskewing

If the scanned page is tilted, I detect its orientation and rotate it to make the text lines horizontal.

Benefit: Correct alignment helps the OCR engine correctly identify characters, words, and text lines.

6. Morphological Processing

I use morphological operations such as **opening and closing** to clean the binary image.

```
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (2, 2))
```

```
clean = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)
```

Opening helps remove small noise, while **closing** can help connect small gaps in characters.

7. Remove Unwanted Regions

Unnecessary borders, lines, stains, or non-text regions should be removed or cropped wherever possible.

This prevents the OCR system from interpreting unwanted regions as text.

8. OCR Recognition

Finally, the preprocessed image is given to an OCR engine such as Tesseract.

```
text = pytesseract.image_to_string(clean)
print(text)
```

Why Each Step Improves Accuracy

Preprocessing Step	Improvement
Grayscale	Simplifies image information
Median filtering	Removes noise
Contrast enhancement	Makes faint text clearer
Thresholding	Separates text from background
Deskewing	Corrects tilted text
Morphological processing	Cleans and repairs characters
Cropping	Removes irrelevant regions
OCR	Converts cleaned image into text

Analytical Justification

The preprocessing methods should be selected according to the type of noise. For example, **median filtering** is suitable for impulse noise, while **adaptive thresholding** is better when the document has uneven illumination. Excessive filtering should be avoided because it may remove important character details. Therefore, the preprocessing pipeline must balance **noise reduction and preservation of character information**.

Conclusion

The proposed pipeline improves OCR by progressively converting a noisy scanned document into a **clean, aligned, high-contrast representation**. Each preprocessing stage addresses a specific problem, resulting in more reliable text recognition and improved OCR accuracy. This demonstrates the independent application of OpenCV concepts required by the assessment.

This version is better for Level 4 because it gives the **method + reason + technical terminology + problem-solving justification + practical OpenCV implementation**, matching the rubric's exemplary expectations.

```

file  edit  format  run  options  window  help
import cv2
import pytesseract

# Tesseract path
pytesseract.pytesseract.tesseract_cmd = r"C:\Program Files\Tesseract-OCR\tesseract.exe"

# Read image
img = cv2.imread(r"C:\Users\DELL\Downloads\python\noisy_document.png.png")

if img is None:
    print("Image not found")
    exit()

# 1. Grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# 2. Noise removal
denoised = cv2.medianBlur(gray, 3)

# 3. Contrast enhancement
enhanced = cv2.equalizeHist(denoised)

# 4. Thresholding
binary = cv2.adaptiveThreshold(
    enhanced, 255,
    cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
    cv2.THRESH_BINARY,
    11, 2
)

# 5. Morphological processing
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (2, 2))
clean = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)

# 6. OCR
text = pytesseract.image_to_string(clean)

print("\n===== OCR OUTPUT =====")
print(text)

# Display results
cv2.imshow("Original Image", img)
cv2.imshow("Grayscale", gray)
cv2.imshow("Noise Removed", denoised)
cv2.imshow("Enhanced", enhanced)
cv2.imshow("Binary Image", binary)
cv2.imshow("Final Clean Image", clean)

```

```

Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python314/CV CO5 AT3.py =

===== OCR OUTPUT =====

>>> = RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python314/CV CO5 AT3.py =

===== OCR OUTPUT =====

|

```

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	
Confidence	10	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	